

SWARM

Industrial IoT Control, Reporting & AI Platform

Project Workflow & Usage Guide

by nanoFarm

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Reference deployment: Tata Steel West Bokaro Biomethanation Plant

Production: <https://app.swarm.co.in>

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1. Project Overview

SWARM is a cloud + edge Industrial IoT platform for biogas, waste-to-energy, and sanitation plants. It combines real-time monitoring, interactive plant HMI, alerts, AI recommendations, predictive maintenance, and multi-plant management.

Core modules

Layer	Module	Purpose
Cloud	Dashboard	Plant health, sensors, alerts, analytics
Cloud	Reports / AI	Automated reports, recommendations, maintenance
Edge	swarm-connect	Device pairing (ESP, PLC, MQTT)
Edge	swarm-hmi	Interactive PFD / P&ID HMI
Edge	swarm-control	Equipment state machines and interlocks
Edge	swarm-io	Sensor and field I/O ingestion
Field	ESP / PLC / Sensors	Physical plant instrumentation

Current phase

- Phase 1-2: Simulation mode - HMI commands update server state only
- Phase 3 (planned): Real PLC / Modbus / MQTT field integration

2. System Architecture Workflow

High-level flow from field to operator:

FIELD DEVICES		EDGE (SWARM Core)		CLOUD (SWARM App)
-----		-----		-----
ESP8266/ESP32	-->	swarm-io / REST ingest	-->	Dashboard
PLC / MQTT	-->	swarm-connect	-->	Analytics
Sensors	-->	swarm-hmi (local PFD)	-->	Plant HMI (React)
		swarm-control (logic)	-->	Alerts / AI / Reports

Technology stack

Layer	Technology
Frontend	React 18, Material UI, Recharts
Backend	Spring Boot 3, Java 17, JWT
Database	MySQL 8
IoT	REST API, MQTT (optional)
HMI runtime	biogas-pfd.html + swarm-hmi.js (iframe)
Deploy	Docker, Railway, Nginx

3. End-to-End Data Flow

Sensor data ingestion

1. Field device (ESP hub, PLC, or gateway) reads sensor values.
2. Data sent via POST /api/iot/data or MQTT topic biopower/sensors/#.
3. Backend stores readings and links them to plant + sensor node.
4. Dashboard and Plant HMI poll latest values (~4-8 seconds).
5. Alert engine checks thresholds; notifications appear in bell icon.

HMI command flow (simulation)

1. Operator clicks block control or master bus in Plant HMI UI.
2. React iframe sends postMessage to swarm-hmi.js.
3. Frontend calls POST /api/hmi/{plantId}/commands or /master.
4. Backend updates hmi_equipment_state and hmi_plant_state tables.
5. UI refreshes; pipe flow animation follows active equipment blocks.

Note: Commands do not reach real PLCs until Phase 3 integration.

4. Plant HMI Workflow

The Plant HMI at /plant-hmi shows the Tata Steel West Bokaro biomethanation process flow diagram with 8 equipment control blocks.

Operator workflow

1. Log in and open Plant HMI from sidebar or /plant-hmi.
2. Plant Admin: Energize master output bus (plant power on).
3. Select equipment block buttons: Belt Conveyor, Crusher, Pre Treatment, Motor, Main Digester, Slurry Storage, Treatment Water, Equalization.
4. Monitor live FIT/PIT/LIT readings in bottom panel (hover on diagram for tag tooltips).
5. Watch animated pipe flow (water, slurry, gas) based on running blocks.
6. Use HMI tabs: Overview, zone pages, alarms, trends, diagnostics, audit.

Block control mapping

UI Block	Tag	Equipment
Belt Conveyor	BC101	Feed belt
Crusher	CH101	Crusher
Pre Treatment Tank	P-101B	Duty feed pump PW
Motor	AG101	Pre-treatment mixer / fan
Main Digester	AG102	Digester agitator
Slurry Storage	P102	Slurry pump
Treatment Water	P104	Treatment water pump
Equalization	P103	Equalization pump

Status colours (bus energized)

Colour	Meaning
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Green	Running
Red	Stopped
Yellow	Fault
Grey	Bus de-energized

5. ESP Sensor Hub Onboarding

Workflow to connect a SWARM MODEL ESP hub (DHT11 + MQ5) to a plant:

1. Admin: set Device Unique ID in firmware and flash ESP once.
2. On site: connect phone to Wi-Fi AP SWARM-Setup-<chipId> (password = Unique ID).
3. Open <http://192.168.4.1/setup> - enter Unique ID and site Wi-Fi credentials.
4. In SWARM app: Connect Device - enter ESP LAN IP + device password.
5. Pair device to plant; verify readings on Dashboard.

Note: Full screenshots: docs/ESP_SENSOR_HUB_MANUAL.md

6. Daily Usage by Role

Super Admin

- Manage users, system settings, MQTT/email thresholds
- Full access to all plants, HMI controls, reports

Plant Admin

- Manage assigned plants and sensor nodes
- Control Plant HMI (energize bus, start/stop equipment)
- Pair ESP devices, acknowledge alerts, generate reports

Operator

- View dashboard, HMI, analytics, alerts
- Cannot send HMI commands or manage plants/users

Typical daily operator flow

1. Open Dashboard - check plant health score and active alerts.
2. Open Plant HMI - verify equipment status and instrument readings.
3. Review Alerts - acknowledge and resolve issues.
4. Check AI Recommendations - follow suggested actions.
5. Generate or download daily/weekly report if required.

7. Development Workflow

Prerequisites

- Java 17+, Node.js 18+, MySQL 8 (or Docker)

Start locally

```
# Terminal 1 - Backend
cd backend
.\mvnw.cmd spring-boot:run -Dspring-boot.run.profiles=dev

# Terminal 2 - Frontend
cd frontend
npm install
npm run dev
```

Key dev URLs

URL	Purpose
http://localhost:3000	React app
http://localhost:3000/plant-hmi	Plant HMI
http://localhost:3000/help	In-app user manual
http://localhost:8080/api	REST API

HMI development

- Edit PFD: frontend/public/hmi/biogas-pfd.html
- Edit logic: frontend/public/hmi/swarm-hmi.js
- React bridge: frontend/src/components/hmi/BiogasPlantDiagram.jsx
- Backend API: backend/.../HmiController.java, HmiService.java

Git workflow

1. Make changes on feature branch or main.
2. Test locally (backend + frontend + HMI).
3. Commit and push to remote (swarm / origin).
4. Railway auto-deploys frontend and backend from main.

8. Deployment Workflow

Docker (local / on-prem)

```
docker-compose up -d
Access: http://localhost
```

Railway (cloud)

Service	Root directory	Notes
MySQL	Railway template	Internal DB host
Backend	backend/	Set PORT, DB, JWT, CORS

Deploy from service root only - not repo root.

Production targets

Host	Role
app.swarm.co.in	SWARM React application
api.swarm.co.in	SWARM backend API (recommended)
swarm.co.in	Marketing website

9. Key URLs & Credentials

Production

- App: <https://app.swarm.co.in>
- Plant HMI: <https://app.swarm.co.in/plant-hmi>

Tata Steel demo (HMI plant)

Role	Email	Password
Plant Admin	tata.admin@tatasteel.com	TataSteel@2026
Operator	tata.operator@tatasteel.com	TataSteel@2026

Local dev defaults

Role	Email	Password
Super Admin	admin@biopower.com	admin123
Plant Admin	manager@biopower.com	manager123
Operator	operator@biopower.com	operator123

10. Project Structure

```

swarm_webapp/
  backend/           Spring Boot API, HMI controllers
  frontend/          React dashboard + Plant HMI page
    public/hmi/       biogas-pfd.html, swarm-hmi.js, P&ID assets
    src/pages/        PlantHmi.jsx, Dashboard.jsx, Help.jsx
  database/          MySQL schema
  docs/              Manuals, checkpoints, deployment guides
  docker-compose.yml Local full-stack deploy

```

Related documentation

- docs/USER_MANUAL.md - full end-user guide
- docs/PLANT_HMI.md - HMI technical reference
- docs/ESP_SENSOR_HUB_MANUAL.md - ESP pairing guide
- docs/RAILWAY_DEPLOYMENT.md - cloud deployment
- docs/PARTNER_API.md - emPOWER partner telemetry API

End of SWARM Project Workflow & Usage Guide